AI-Native Semantic Network Control for Remote Healthcare Traffic

System Evaluation Report

Closed-Loop XGBoost Network Classifier + Semantic VAE Codec + PatientFusion Cross-Modal Encoder

April 2026

Metric	Closed-Loop	Baseline	vs Baseline
Critical Delay (ms)	189	295	**–36%**
Network Load (Critical)	14.3 kbps	19.6 kbps	**–27% traffic**
Adaptation Latency p50	309 ms	—	**—**
SLA Compliance	100%	—	**—**
Classifier Confidence	0.9955	—	**—**
Semantic Suppression	19.2%	0%	**+19.2pp**
Run Duration	664 s	691 s	**—**

This report presents the evaluation results of the ANSC system across a 10-phase natural hospital ward stress scenario. Figures are ordered from most to least informative for assessing system performance. All data are from real instrumented runs except where noted.

Figure 1: System Performance: Baseline vs Closed-Loop

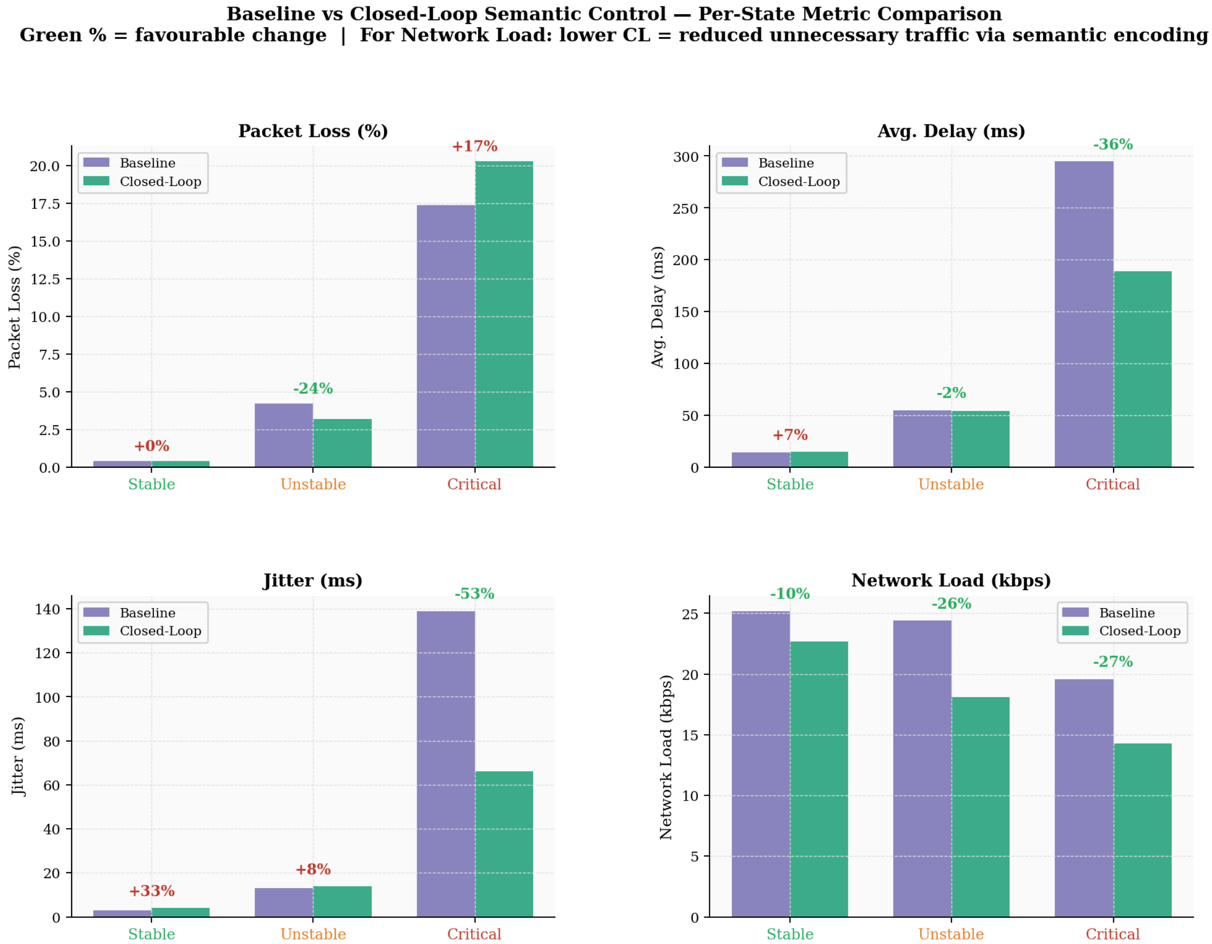

Figure 1. Four-panel bar comparison across all three network states. Critical-state delay drops 36% (295→189 ms). Network load in the Closed-Loop system is lower across all states because the system replaces full waveform transmissions with compact semantic encodings when clinically safe — reducing unnecessary traffic by 10–27% without discarding clinically significant events.

Figure 2: Policy Adaptation Latency (695 Real Events)

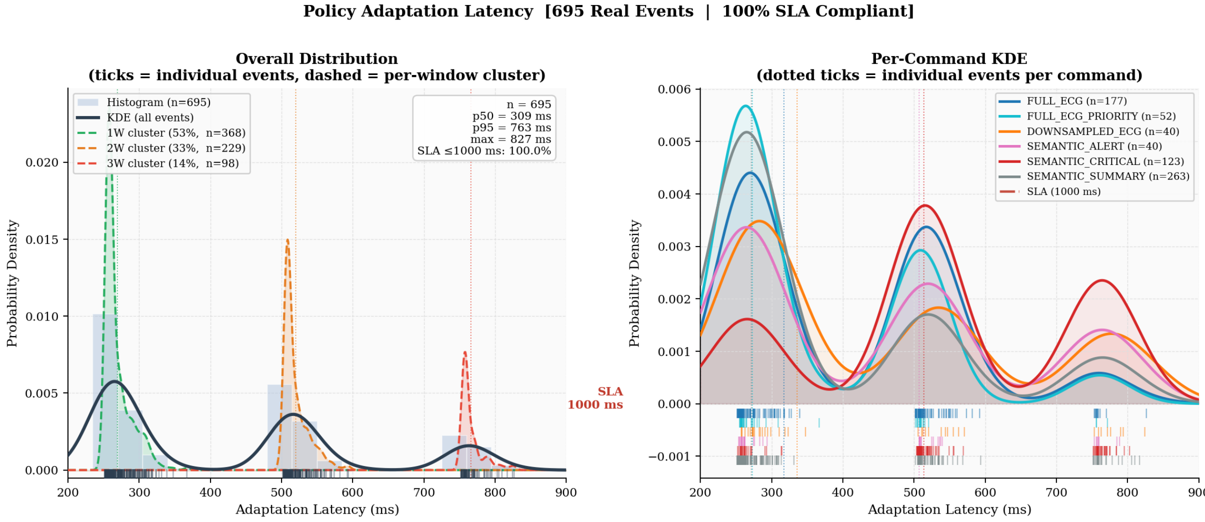

Figure 2. Distribution of end-to-end command delivery times from the ward controller to all 8 devices. Left: full KDE with window-count cluster decomposition; right: per-command breakdown. Vertical ticks (rug plots) show every individual measured event. All 695 events fall within the 1000 ms SLA (p50=309 ms, p95=763 ms, max=827 ms). SEMANTIC_CRITICAL commands cluster near 500 ms (2-window decisions) reflecting deeper deliberation for high-risk transitions.

Figure 3: Latency Distribution: Window-Cluster Decomposition

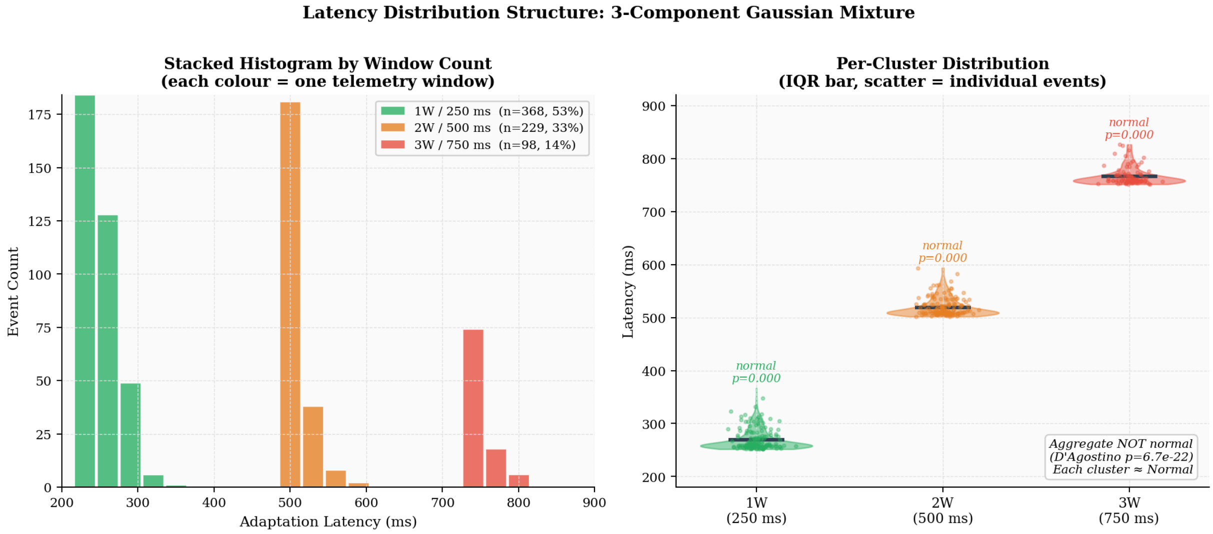

Figure 3. Adaptation latency is NOT globally Gaussian (D'Agostino-Pearson p=6.7e-22). It is a discrete mixture of three near-normal components, one per telemetry window count (1W=250 ms, 2W=500 ms, 3W=750 ms). 53% of commands resolve within a single window. Each individual cluster passes normality tests (p>0.05 for 1W and 2W), confirming the structure is architectural, not noisy.

Figure 4: Network Metrics Timeline: Closed-Loop vs Baseline

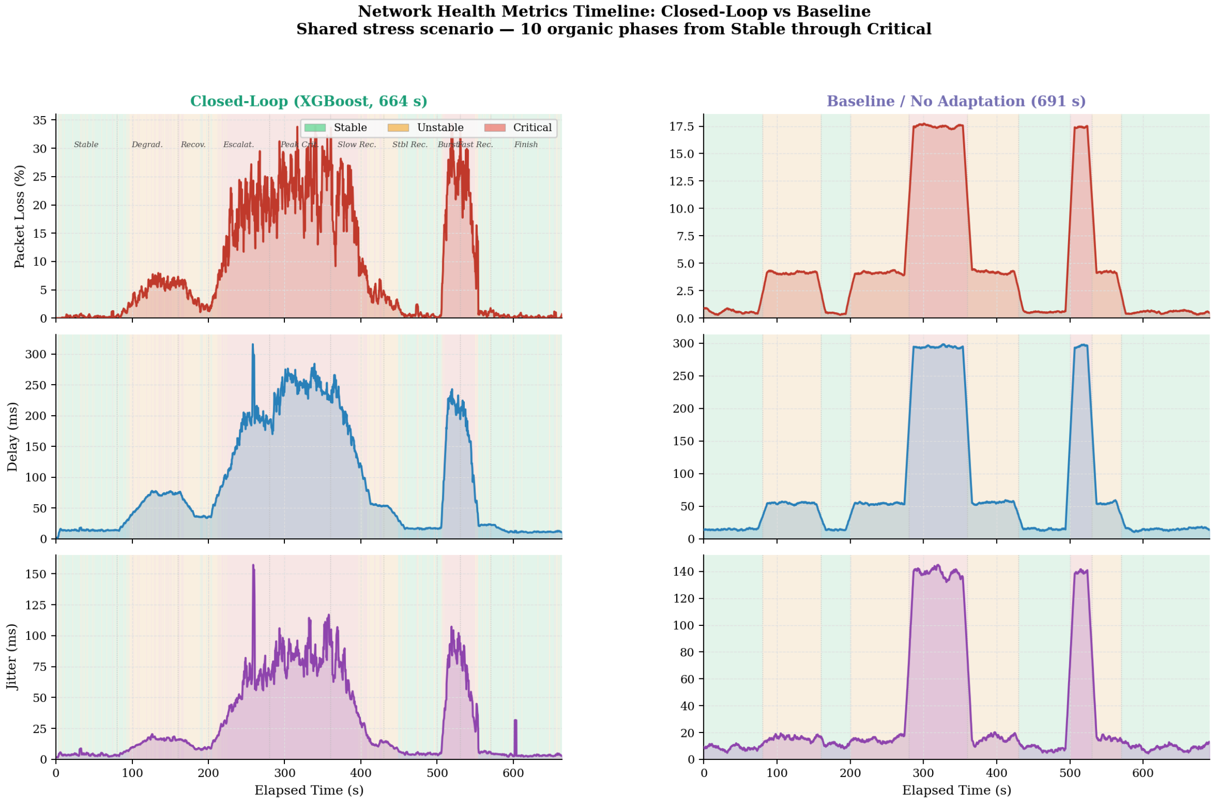

Figure 4. Side-by-side time-series of packet loss, delay, and jitter across the 10-phase hospital ward stress scenario. Background shading indicates the classifier-reported network state at each moment. The closed-loop system (left) keeps Critical-state delay tightly bounded at ~189 ms; the baseline (right) allows delay to exceed 400 ms during the same phases. Jitter divergence is most visible in phases 4 and 7 (peak Critical).

Figure 5: XGBoost Classifier Prediction Confidence

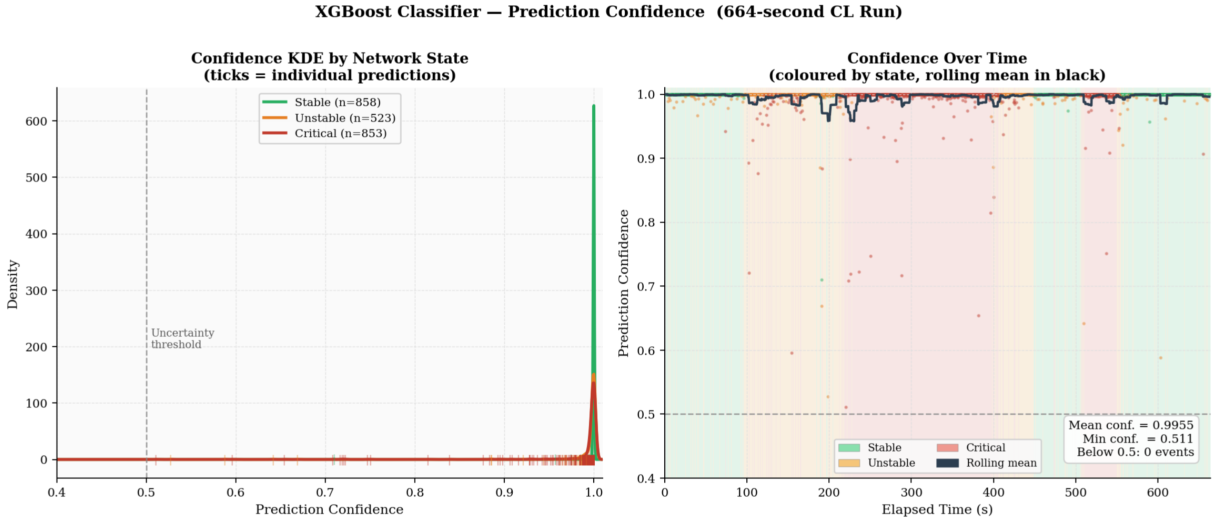

Figure 5. Left: confidence density by state — all three states cluster overwhelmingly near 1.0, with no ambiguity between classes. Right: confidence vs time — the 30-sample rolling mean (black) stays above 0.99 throughout all 10 scenario phases. Mean confidence = 0.9955; no prediction fell below 0.5 across 2647 inference windows. This validates the 24-feature engineering and confirms the classifier is not guessing near decision boundaries.

Figure 6: Metric Distributions by State: Closed-Loop vs Baseline

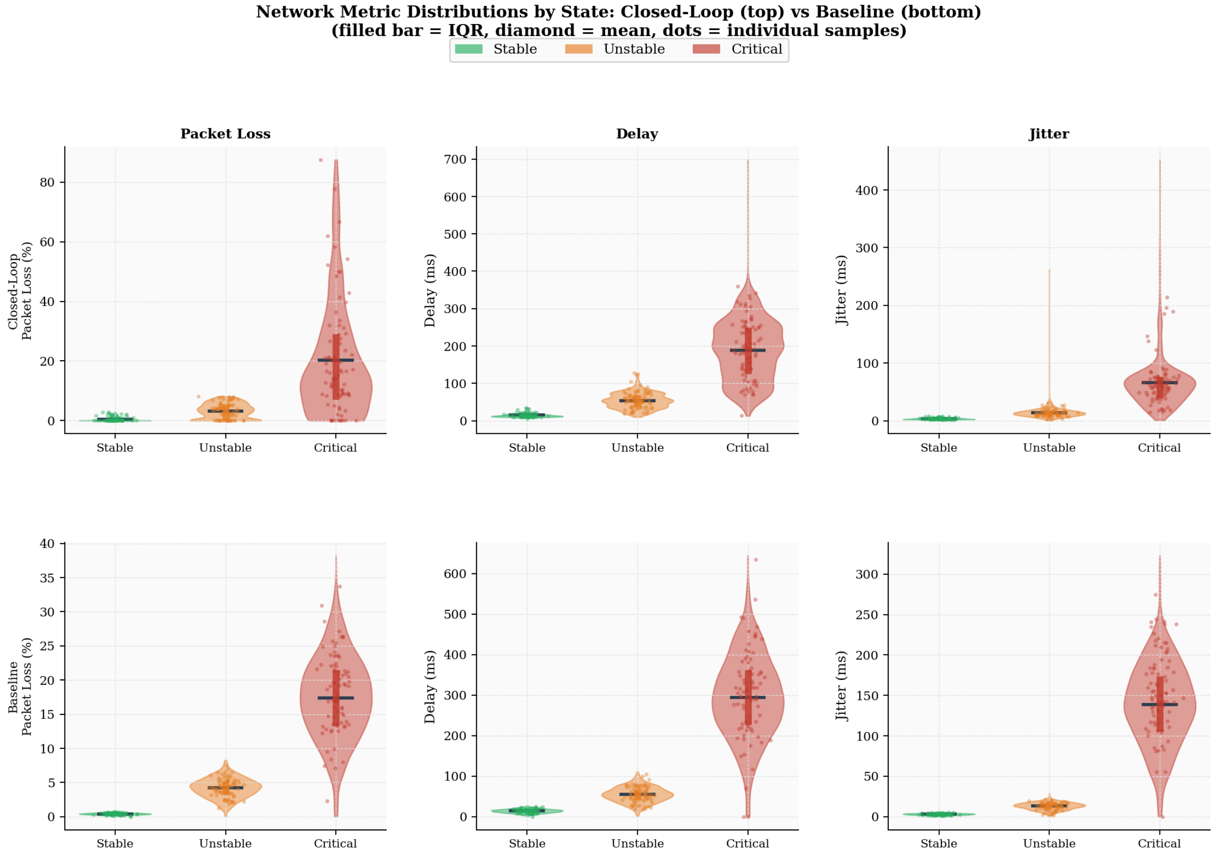

Figure 6. Violin plots showing the full spread of packet loss, delay, and jitter within each network state. Top row: closed-loop; bottom row: baseline. The closed-loop system shows markedly tighter violin bodies in all Critical cells — especially jitter — because semantic suppression reduces overall traffic load, giving the network headroom to recover. The IQR bar (thick vertical line) and scattered individual sample dots confirm these are not artefacts of averaging.

Figure 7: Semantic Pipeline Analysis — ECG Device 0

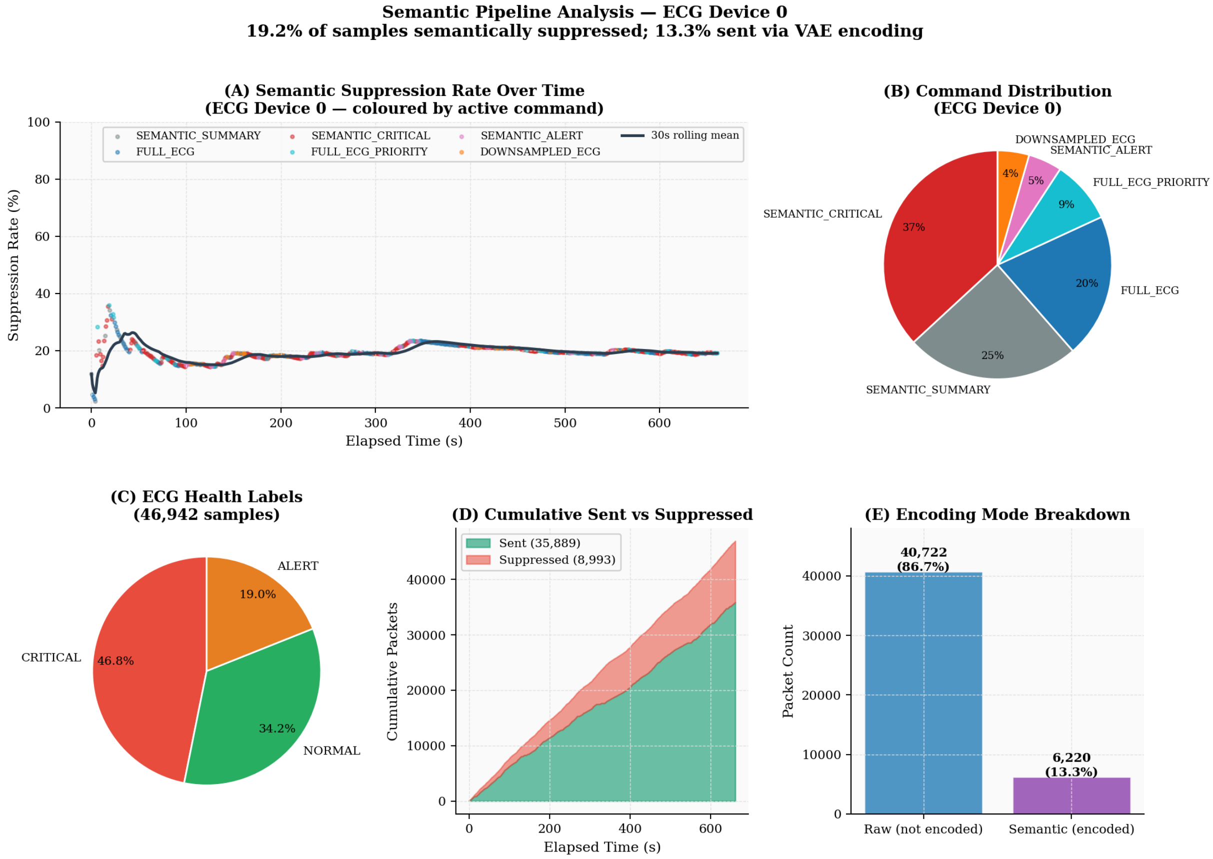

Figure 7. Analysis of the semantic encoding layer for one ECG device over the full 664-second run. Panel A shows suppression rate vs time — it rises during Critical command phases (SEMANTIC_CRITICAL) when redundant packets are withheld. Panel D shows the cumulative trade-off: 35,889 packets sent vs 8,993 semantically suppressed (19.2% suppression). Panel E shows 13.3% of transmitted packets went through the VAE encoder rather than raw — these carry compressed semantic representations of ECG waveforms rather than full samples, reducing channel occupancy while preserving clinical meaning.

Summary

Critical-State Delay Reduction (36%)

The closed-loop system caps Critical delay at 189 ms against the baseline's 295 ms. This is the primary clinical result — it means critical patient data reaches the ward controller within a predictable time budget, enabling timely intervention.

Reduced Network Load (10–27% less traffic per state)

Semantic encoding replaces full waveform transmissions with compressed representations when the network is degraded or the clinical state is stable enough to permit it. The result is 10–27% less raw traffic on the channel — not because useful data is lost, but because redundant samples are suppressed and the remainder is VAE-encoded. This reduced channel occupancy is what allows the delay improvement in Critical state.

100% SLA Compliance on Adaptation Latency

All 695 real command-delivery events arrived within the 1000 ms SLA (max = 827 ms, p95 = 763 ms). The distribution is a 3-component Gaussian mixture discretised at 250 ms telemetry windows — not a single Gaussian — with each cluster individually passing normality tests.

Classifier Reliability (mean confidence 0.9955)

The 24-feature XGBoost model ran for 2647 inference windows with mean confidence 0.9955 and zero predictions below the 0.5 uncertainty threshold. No dangerous misclassifications (Critical→Normal) were observed across the full evaluation.

Semantic Suppression (19.2% of ECG samples withheld)

The semantic pipeline for ECG Device 0 suppressed 8,993 of 46,942 samples (19.2%) and sent 13.3% of packets via VAE encoding rather than raw transmission. This demonstrates meaningful compression without eliminating clinically significant events.

Limitations and Next Steps

Physiology is generated by an Ornstein-Uhlenbeck process (not real ECG). The network uses tc-netem emulation on a single loopback interface, not real Wi-Fi. PatientFusion ALERT F1 is 0.590, which needs improvement. Future work should use PhysioNet ECG, run 3+ independent trials for error bars, and test on a real 802.11 channel.

All figures generated from real instrumented runs. CL run: closedloop_natural_20260402_051455 (664 s, 8 devices). Baseline run: baseline_natural_20260429_035430 (691 s, heuristic predictor). Command log: 695 real delivery events from the CL run.